

AI FITTRACK

PERSONALIZED FITNESS RECOMMENDATIONS POWERED BY AI

PHASE 1 – BRAINSTORMING & IDEATION

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Phase boundary

The Brainstorming & Ideation phase establishes why the project exists, what problem it addresses, what solution was selected, and what the initial scope contains. Detailed implementation, API testing and final demonstration are intentionally handled in later project phases.

1. PHASE OVERVIEW

The Brainstorming & Ideation phase establishes the foundation for AI FitTrack.

The assigned project describes a backend REST API for workout tracking and AI-assisted fitness guidance. The ideation stage therefore focuses on the practical problem of maintaining structured workout information, securing user access, retrieving workout records efficiently and providing personalized fitness guidance through an AI service.

The project is intentionally treated as a backend application. The supplied implementation uses Node.js, Express.js, MongoDB, Mongoose, JWT authentication, bcryptjs and Google Gemini. The project can be consumed by an API client and can conceptually support a web or mobile client, but the supplied implementation itself is the backend API.

2. PROJECT IDEA AND BACKGROUND

Fitness applications require a reliable mechanism for recording exercise activities and presenting meaningful information to users. Workout information may include the type of activity, duration, calories burned and date of exercise. When these records are maintained manually, organizing historical activities and searching previous sessions can become difficult.

The assigned AI FitTrack concept addresses this problem through a centralized backend service. Instead of treating workout information as unstructured personal notes, the system represents it as structured records associated with an authenticated user. This creates a consistent API boundary for storing, retrieving, searching, updating and deleting workout information.

The project extends the workout-recording concept with AI-assisted guidance. Two AI functions are defined in the assigned specification: workout recommendations based on age, fitness goal and experience level, and fitness insights based on workout statistics such as total workouts, average duration and total calories burned.

The resulting idea is therefore a secure and modular backend that combines workout data management with AI-assisted fitness guidance.

3. PROBLEM IDENTIFICATION

The ideation process identifies the following practical problems addressed by the assigned project:

- Workout information is difficult to organize manually. Repeatedly maintained records can become difficult to retrieve and manage as the number of sessions increases.
- Workout history requires structured storage. A consistent data model is needed so that workout information can be stored and retrieved through predictable operations.
- Personal fitness information requires controlled access. Authentication is needed so that protected operations can be associated with an authenticated user.
- Searching previous workouts can be inconvenient. Users need mechanisms to search records by workout name, category and date.
- Personalized guidance is not inherent in basic workout storage. The system therefore includes an AI recommendation capability.
- Workout statistics can require interpretation. The AI fitness-insights function provides an additional mechanism for interpreting supplied workout statistics.

These problems form the basis for the proposed system without expanding the assigned scope into unrelated frontend, administration or additional fitness-management modules.

4. IDEATION CONSIDERATIONS

The project concept can be represented as a set of design considerations that connect the identified problems to the selected solution.

Problem / Need	Ideation Direction	Selected Project Response
Organized workout records	Use structured backend data models	Workout model stored through MongoDB/Mongoose
Controlled access	Authenticate users before protected operations	JWT-based authentication and protected routes
Secure passwords	Hash passwords before storage	bcryptjs password hashing
Workout retrieval	Provide standard REST operations	GET, POST, PUT and DELETE endpoints
Workout discovery	Provide searchable workout records	Search by name, category and date
Personalized guidance	Connect the backend to an AI service	Google Gemini integration
Fitness performance interpretation	Provide a separate insights operation	AI fitness insights endpoint
Maintainable implementation	Separate application responsibilities	Routes, controllers, middleware, models and services

5. PROPOSED AI FITTRACK CONCEPT

AI FitTrack is proposed as a secure, modular backend REST API for personalized fitness recommendations and workout management. The authenticated user interacts with the API through an API client. Express.js receives the request and routes it to the appropriate controller. Protected operations use JWT authentication middleware. Controllers interact with Mongoose models for database operations and invoke the dedicated Gemini service for AI functionality.

The project maintains two primary data entities in the supplied implementation: User and Workout. A user may have multiple workout records. This simple data model keeps the initial system focused on the assigned workout-management and AI-assistance requirements.

The proposed concept is organized into the following major capabilities:

Capability	Concept
Authentication	Register, login and protected profile access using JWT authentication.
Workout Management	Create, retrieve, update and delete workout records belonging to the authenticated user.
Workout Search	Search workout records using name, category or date.
AI Recommendation	Generate personalized workout guidance using age, fitness goal and experience level.
AI Fitness Insights	Generate insights from total workouts, average duration and total calorie
Data Persistence	Store users and workouts using MongoDB with Mongoose models.
Application Structure	Separate routes, controllers, middleware, models, configuration and AI service responsibilities.

The concept is intentionally limited to the supplied backend scope. A separate React interface, administrator module and additional fitness collections are not treated as current project features.

6. PROJECT OBJECTIVES

The ideation phase establishes the following objectives for the AI FitTrack project:

1. Implement secure user registration, login and profile retrieval.
2. Protect authenticated API routes using JWT middleware.
3. Hash user passwords using bcryptjs before storage.
4. Provide complete workout create, read, update and delete operations.
5. Provide workout search by name, category and date.
6. Integrate Google Gemini for AI-powered workout recommendations.
7. Integrate Google Gemini for fitness insights based on workout statistics.
8. Maintain a modular MVC-oriented backend structure.
9. Provide consistent JSON-based REST API responses.
10. Produce project evidence suitable for GitHub and Skill Wallet review.

7. INITIAL PROJECT SCOPE

In Scope	Out of Current Scope / Future Extension
Node.js and Express.js backend	Separate React frontend
MongoDB and Mongoose	Administrator dashboard
User registration, login and profile	Advanced role-based administration
JWT authentication and bcryptjs password hashing	Notifications module
Workout CRUD operations	Additional fitness collections not present in the implementation
Workout search by name, category and date	Progress/goal/history collections beyond the supplied User and Workout models
Gemini workout recommendation	Production deployment and monitoring
Gemini fitness insights	Unverified third-party integrations
REST API and modular backend structure	Features not represented in the assigned project scope

8. CORE FUNCTIONAL CONCEPT

The selected concept can be summarized as a sequence from secure user access to workout management and AI assistance.

Stage	Conceptual Flow
1. User Access	User registers or logs in to obtain authenticated access.
2. Authentication	JWT middleware protects secured operations.
3. Workout Data	Authenticated user creates and manages workout records.
4. Search	Workout records can be retrieved by name, category or date.
5. AI Recommend	User fitness inputs are submitted to the recommendation endpoint.
6. AI Processing	Gemini service sends the request to Google Gemini.
7. AI Response	The API returns generated recommendation or fitness insight information.

9. EXPECTED BENEFITS

The expected benefits of the proposed concept are derived from the problems identified during ideation and the functions specified for AI FitTrack.

- Centralized workout management: workout information is maintained as structured records in a database.
- Authenticated access: protected operations are associated with the authenticated user.
- Efficient retrieval: workout records can be retrieved and searched through REST endpoints.
- AI-assisted guidance: users can request personalized work out recommendations.
- Fitness interpretation: users can submit workout statistics for AI-generated fitness insights.
- Modular maintainability: separated routes, controllers, middleware, models and services make the backend easier to understand and test.
- Extensible foundation: the modular API can be extended in future phases if additional requirements are formally approved.

10. TEAM FORMATION AND INITIAL RESPONSIBILITIES

No.	Team Member	Initial Responsibility
1.	CHANDRA SEKAR G	Team Leader – project integration, backend coordination, AI integration, final testing
2.	CHANDRU G T	Backend Development and API Integration – route configuration, controller implementation
3.	CHANDRU M	MongoDB Configuration and Database Design – MongoDB Atlas, User/Workout schemas
4.	BOOPATHI B L	API Testing and Verification – authentication, workout CRUD/search, AI endpoints, testing evidence
5.	CHANDAN KUMAR B	Architecture, Documentation and Execution Evidence – diagrams, project structure documentation, execution evidence

11. IDEATION OUTCOME

The Brainstorming & Ideation phase results in a clearly bounded project concept: AI FitTrack will be developed as a backend REST API that combines authenticated workout management with AI-assisted fitness guidance.

The ideation outcome establishes the following baseline:

Decision Area	Baseline Outcome
Project	AI FitTrack – Personalized Fitness Recommendations Powered by AI
Primary Application	Backend REST API
Primary User	Authenticated user
Database	MongoDB with Mongoose
Primary Data Models	User and Workout
Authentication	JWT + bcryptjs
AI Service	Google Gemini through a dedicated service module
Workout Functions	Create, retrieve, search, update and delete
AI Functions	Workout recommendation and fitness insights
Architecture Style	Modular MVC-oriented backend
Evidence Direction	Phase-wise GitHub submission and Skill Wallet evidence

12. PHASE 1 CONCLUSION

The Brainstorming & Ideation phase establishes the rationale, concept, objectives and boundaries of AI FitTrack. The project addresses structured workout management, authenticated access, search and AI-assisted fitness guidance through a modular backend REST API.

The phase deliberately avoids claiming unsupported implementation features. The defined concept provides a stable basis for requirement analysis, system design, project planning, development, testing, documentation and final demonstration.

Phase 1 Status: Brainstorming & Ideation documented.